GW170817 confines the vestigial-second-sound graviton identification to one part in 10^{15} of its natural range

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We constrain our assumed leading-order graviton speed $c_{\rm GW} = c\sqrt{\chi_{\rm vest}}$. The conservative GW170817 bound $|\Delta c/c| \le \tau = 3\times10^{-15}$ confines the nonnegative susceptibility to $[(1-\tau)^2, (1+\tau)^2]$, of width 4τ. This interval contains $\chi_{\rm vest} = 1$ and is over fourteen orders of magnitude narrower than our adopted range $[0.1, 10]$. The mode and range are not derived from microscopic dynamics; the constraint applies to this kinematic model, not to emergent gravity generally.

INTRODUCTION

In an emergent-gravity model, the spin-2 graviton could be associated with second sound in a vestigial fluid. Volovik proposes a second-sound identification in Refs. [3, 4]. His earlier counting [2] associates six Goldstone modes with the spin connection in the Akama–Diakonov–Wetterich formulation [5–9]. We test the hypothesis that the two tensor helicities $h_+, h_\times$ obey the second-sound wave equation. Establishing the propagating modes and their dynamics requires a microscopic derivation.

Whose claim is Eq. (1). The *identification* is Volovik's; the susceptibility-dependent speed is ours. He advances it in [3, 4] — not in the vestigial-gravity note [1], which concerns the two-step transition — and on his own de Sitter two-fluid background the Landau formula yields $s_2 = c$ *exactly*, independent of H: a structural consequence of the stiff-matter equation of state $w_n = 1$, not a tunable parameter, and hence GW170817-compatible at leading order. He leaves the spin assignment open, describing a scalar mode; the spin-2 assignment above comes from the ADW Goldstone counting and is likewise ours. Transplanting the identification onto the ADW substrate gives the following conditional speed law. Its $\chi_{\rm vest}$ is a dimensionless parameter. The metric-channel model below supplies a conditional RPA relation, and a microscopic matching calculation would be needed to fix $\chi_{\rm vest}$:

$$c_{\rm GW}^2(\chi_{\rm vest}) = c^2 \cdot \chi_{\rm vest}, \qquad c_{\rm GW}(\chi_{\rm vest}) = c\sqrt{\chi_{\rm vest}}, \quad (1)$$

where $\chi_{\rm vest}$ represents a normalized metric-channel response. It is a *free parameter* here, and constraining it is what this Letter does; the random-phase-approximation form our substrate models it with is set out in `VestigialSusceptibility.lean`, but no value of $\chi_{\rm vest}$ is derived there or anywhere else in the project. The underlying lattice-fermion framework is the one whose unitarity Vergeles establishes [10]. The identification is appealing in its brevity. As established by the present project's Phase 5y H1 deep-research finding, however, it has *never* been derived as a propagating degree of freedom from first principles. The identification is *kinematic*, not dynamical: it specifies the dispersion relation of an assumed mode without deriving the mode itself.

The 2017 binary-neutron-star merger GW170817 [11] placed an extraordinarily tight bound on $c_{\rm GW}$: comparing the gravitational-wave arrival time at LIGO/Virgo against the gamma-ray-burst signal at Fermi/INTEGRAL, the multi-messenger collaboration constrained $-3\times10^{-15} \le \Delta c/c \le +7\times10^{-16}$. Conservatively two-sided, $|\Delta c/c| \le \tau$ with $\tau = 3\times10^{-15}$.

This Letter formalizes Eq. (1), places it head-to-head against GW170817, and reports the result. The conclusion is sharp: our assumed leading-order speed law satisfies the GW170817 bound only on a sub-interval of the natural range $\chi_{\rm vest} \in [0.1, 10]$ narrower than that range by more than 14 orders of magnitude. The GW170817-compatible window $[(1-\tau)^2, (1+\tau)^2]$ has width 4τ in $\chi_{\rm vest}$ and sits strictly interior to our adopted range $[0.1, 10]$ about the anchor $\chi_{\rm vest} = 1$. Its relative width is $4\tau/9.9 \approx 1.2\times10^{-15}$; a probability mass additionally requires a specified prior. The result is a fine-tuning constraint on one specific kinematic identification, not a refutation of emergent gravity nor of the ADW program; we discuss three Phase 6 recovery paths in Sec. .

Relation to the 2017 constraint sweep. Using GW170817 to eliminate a gravity model is not new. Within months of the detection, four back-to-back Letters established what the bound kills: Baker *et al.* [12] covered general scalar-tensor and vector-tensor theories and bounded the graviton mass in bimetric gravity; Creminelli and Vernizzi [13] showed that avoiding tuning requires $c_{\rm GW}$ to be unaffected on *nearby* cosmological solutions, collapsing quartic and quintic beyond-Horndeski theories to a single function; Sakstein and Jain [14] combined the bound with cluster and M87 strong-equivalence-principle data to render the quartic galileon cosmologically irrelevant; and Ezquiaga and Zumalacárregui [15] extended the conclusion to Einstein-aether, Hořava, Generalized Proca and TeVeS.

The present result is not a further entry in that list, and the difference is structural rather than one of degree. Every theory the 2017 sweep reached has a Lagrangian from which a varying $c_{\rm GW}$ is *derived*; the constraint therefore lands on operator coefficients, and each

Letter reports which coefficients survive. The vestigial-second-sound identification has no such Lagrangian. As Sec. notes, $c_{\rm GW} = c\sqrt{\chi_{\rm vest}}$ is asserted kinematically: the mode is assumed rather than derived, so there are no operator coefficients for the bound to constrain and nothing to tune. What the bound constrains is $\chi_{\rm vest}$: the compatible interval has roughly one part in 10^{15} of the width of our adopted order-unity window. This is an interval-width comparison, not a microscopic prediction of $\chi_{\rm vest}$ or a probability statement independent of a prior. The 2017 sweep constrained *theories*; this constrains an *identification*, which is why it is stated as a ratio of two interval widths rather than as a surviving region of parameter space.

A corollary worth stating plainly: because the argument never uses a Lagrangian, it is insensitive to the operator-level escape routes those Letters catalogue. Conformal rescaling, coefficient compensation and restriction to Horndeski's simplest terms all act on terms that the kinematic identification does not have. The three recovery paths in Sec. can supply a mechanism fixing $\chi_{\rm vest}$ within the compatible interval, or modify the identification or its leading-order assumptions. A fixed-point mechanism can preserve the speed law; the finite-tolerance constraint alone derives none of these mechanisms.

FORMAL SETUP AND RESULT

Vestigial-susceptibility natural range. Our substrate models the vestigial-phase metric-channel susceptibility by the RPA *resummation* $\chi_{\rm RPA} = \chi_0/(1-\gamma_\star\chi_0)$ in the standard Hertz–Millis convention, where χ_0 is the bubble integral — the two are distinct, and $\chi_{\rm RPA}$ is not itself a bubble integral. `VestigialSusceptibility.lean` formalizes that closed form and its stability properties; it evaluates neither χ_0 nor $\gamma_\star$, so it fixes no numerical $\chi_{\rm vest}$, which is why $\chi_{\rm vest}$ enters Eq. (1) as a free parameter. The natural range $\chi_{\rm vest} \in [0.1, 10]$ is a *project-adopted* naturalness window: the one-loop bubble normalization is $\mathcal{O}(\Lambda_{\rm UV}^2/16\pi^2)$ by dimensional analysis, and we take the $\pm$1-decade window about the resulting unit-normalized central value. We emphasize the provenance: the window is a modeling choice of this project, not a published derivation: the Vergeles lattice analysis [10] establishes unitarity of the underlying Einstein–Cartan–Palatini lattice theory but does not itself compute $\chi_{\rm vest}$ or assign it a range. The window matches the linearized-EFE wave's ADW microscopic-coefficient $\alpha_{\rm ADW} \in [0.1, 10]$ natural range [16], which we treat as the project-internal anchor for the order-of-magnitude prior.

The leading-order $c_{\rm GW}$. Equation (1) ships in `GravitationalWaves.lean` as `c_GW(chi, c) := c * Real.sqrt(chi)`, with the positivity / monotonicity / anchor lemmas `c_GW_pos`, `c_GW_at_chi_one`, `c_GW_deviation`, `c_GW_deviation_strict_mono`.

The GW170817 correctness-push. Define $\Delta c/c(\chi_{\rm vest}) = \sqrt{\chi_{\rm vest}} - 1$ and the predicate $\texttt{LigoSatisfied}(\delta) \equiv |\delta| \le \tau$. The correctness-push biconditional is

Theorem 1 (`c_GW_match_iff_chi_close_to_one`). *For $\chi_{\rm vest} \ge 0$, $\texttt{LigoSatisfied}(\Delta c/c(\chi_{\rm vest})) \iff \chi_{\rm vest} \in [(1-\tau)^2, (1+\tau)^2]$.*

The proof is a $\sqrt{\cdot}$/squaring round-trip via `Real.sqrt_le_sqrt` and `Real.sqrt_sq`, with `nlinarith` supplied the hint $\sqrt{\chi_{\rm vest}}^2 = \chi_{\rm vest}$. The compatible $\chi_{\rm vest}$ window has width $\sim 4\tau = 1.2 \times 10^{-14}$.

Natural-range fine-tuning. The natural range $[0.1, 10]$ has width 9.9 (`natural_range_width_eq`); the GW170817-compatible window has width exactly $4\tau = 1.2 \times 10^{-14}$ (`ligo_window_width_eq`) and is nested strictly inside it (`ligo_window_subset_natural_range`), not disjoint from it. Scaling the window up by 10^{14} still leaves it the narrower (`ligo_window_width_lt_natural_range_width_by_1e14`): that ratio, not a disjointness, is the fine-tuning claim. Two endpoint-falsifier theorems sharpen the exclusion of the extremes.

Theorem 2 (`natural_lower_violates_ligo`). *At $\chi_{\rm vest} = 0.1$, $\Delta c/c = \sqrt{0.1} - 1 \approx -0.6838$, hence $|\Delta c/c|/\tau \approx 2.28 \times 10^{14}$ and* `LigoSatisfied` *is false.*

Theorem 3 (`natural_upper_violates_ligo`). *At $\chi_{\rm vest} = 10$, $\Delta c/c = \sqrt{10} - 1 \approx +2.1623$, hence $|\Delta c/c|/\tau \approx 7.21 \times 10^{14}$ and* `LigoSatisfied` *is false.*

The proofs use $\sqrt{1/10} \le \sqrt{1/4} = 1/2$ and $\sqrt{10} \ge \sqrt{4} = 2$, both via `Real.sqrt_le_sqrt` and `Real.sqrt_sq`, closed by `linarith`. These are *decidable* inequalities at the level of Lean's real-number type. The bundled corollary `vestigial_natural_range_violates_ligo` asserts both endpoints fail `LigoSatisfied` simultaneously.

The identification hypothesis itself is refuted pointwise rather than only at the observable: `H_VestigialModeIsGraviton_fails_at_natural_lower` and `H_VestigialModeIsGraviton_fails_at_natural_upper` refute it at the two endpoints, and `H_VestigialModeIsGraviton_fails_at_zero` and `H_VestigialModeIsGraviton_fails_at_quarter` refute it at $\chi_{\rm vest} \in \{0, 1/4\}$. Zero lies outside the adopted range, whereas 1/4 is an interior point. These falsifiers show failures away from the two natural-range endpoints, without excluding the compatible interval. `ligo_satisfied_at_chi_one` supplies the compatible anchor $\chi_{\rm vest} = 1$. The exact-zero statement `c_GW_deviation_zero_iff_chi_one` characterizes vanishing deviation, not finite-tolerance compatibility: the latter holds on the entire interval in Theorem 1.

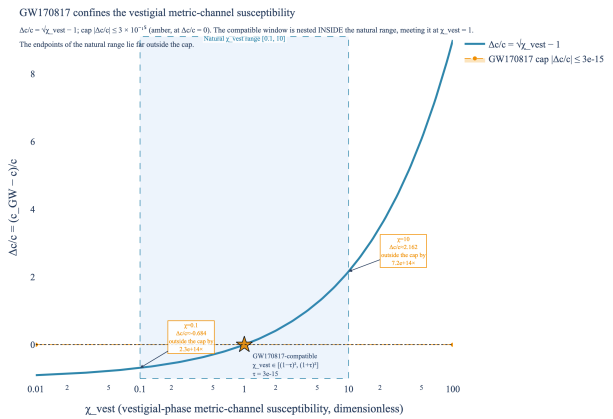


FIG. 1. **Headline result.** The $\Delta c/c = \sqrt{\chi_{\rm vest}} - 1$ deviation across the natural $\chi_{\rm vest} \in [0.1, 10]$ range (pale blue band, dashed blue border) versus the GW170817 [11] multi-messenger cap $|\Delta c/c| \leq 3 \times 10^{-15}$ (amber, drawn at $\Delta c/c = 0$ — the cap's own width is far below one pixel at this scale). Over the natural range $\Delta c/c$ sweeps $[-0.68, +2.16]$, exceeding the cap by $\sim 10^{14}\times$ *at the endpoints* (orange callouts) and vanishing at $\chi_{\rm vest} = 1$. The compatible $\chi_{\rm vest}$ window $[(1-\tau)^2, (1+\tau)^2]$ has width $4\tau = 1.2 \times 10^{-14}$, compared with the adopted range's width 9.9. Its finite extent is unresolved at this scale; the gold star marks its anchor $\chi_{\rm vest} = 1$, not a singleton compatible set. The window sits *inside* the band: the datum confines $\chi_{\rm vest}$, it does not exclude the range. Lean witnesses: `ligo_confines_chi_vest_within_natural_range`, `ligo_window_subset_natural_range`, `vestigial_natural_range_violates_ligo`.

Tracked-hypothesis bundle. Per the project's discipline for non-derived identifications, we bundle the structural properties of "the vestigial mode is the graviton" into a Lean Prop with three recorded conjuncts:

$$\begin{aligned} &\texttt{H_VestigialModeIsGraviton}(\chi_{\rm vest}) \\ &\quad \equiv \chi_{\rm vest} > 0 \\ &\qquad \wedge \texttt{LigoSatisfied}(\Delta c/c(\chi_{\rm vest})) \\ &\qquad \wedge |\Delta c/c(\chi_{\rm vest})| < 1/2, \end{aligned}$$

demanding positivity (P1), GW170817 compatibility (P2), and sub-half-deviation $|\Delta c/c| < 1/2$ (P3'). These clauses are not independent at the fixed tolerance $\tau = 3 \times 10^{-15}$: P2 already implies P3' because $\tau < 1/2$, and the compatible interval in Theorem 1 is strictly positive, supplying P1 throughout that interval. The recorded Prop retains all three clauses; this observation does not change its definition.

The bundle is satisfied at $\chi_{\rm vest} = 1$ (`H_VestigialModeIsGraviton_at_one`). Four explicit falsifier theorems establish that it is not universally satisfied: `*_fails_at_natural_lower` and `*_fails_at_natural_upper` use failure of P2; `*_fails_at_zero` uses failure of P1; `*_fails_at_quarter` uses failure of P3'. At $\chi_{\rm vest} = 1/4$, $\Delta c/c = -1/2$ fails both P2 and P3', while satisfying P1. It therefore shows that P1 alone does not imply P3', not that P3' is independent of the other two clauses. Likewise these proofs do not isolate one failing clause while requiring all others to hold. On the nonnegative physical domain, the bundle's compatible set is precisely the finite interval in Theorem 1, of width $4\tau = 1.2 \times 10^{-14}$. Its probability mass depends on the prior; under a uniform prior on $[0.1, 10]$ it is $4\tau/9.9$.

Phase 5y H1 caveat as a Lean endpoint theorem. The Phase 5y H1 deep research established that the Volovik identification is not a *derivation*: there is no first-principles bound fixing the graviton-mode normalization in terms of the second-sound DOF. We formalize this caveat as the endpoint theorem `natural_endpoints_violate_ligo_window`: for $\chi_{\rm vest}$ at either natural-range endpoint $\{0.1, 10\}$, no c-rescaling gives a GW170817-compatible $\Delta c/c$. The frame-independent form `natural_endpoints_violate_ligo_for_every_c` quantifies over all $c > 0$ explicitly, resting on `c_GW_relative_deviation_eq`. Both concern the two endpoints only; neither excludes interior values, and $\chi_{\rm vest} = 1$ satisfies the cap (`ligo_satisfied_at_chi_one`).

DISCUSSION

What this constrains. The conservative symmetric cap restricts the assumed leading-order identification to $\chi_{\rm vest} \in [(1-\tau)^2, (1+\tau)^2]$. Expanding the endpoints gives $1 - 2\tau + \tau^2$ and $1 + 2\tau + \tau^2$: the susceptibility offsets are approximately $\pm 6 \times 10^{-15}$, whereas the speed-deviation tolerance is $\pm 3 \times 10^{-15}$. Exact luminality occurs at $\chi_{\rm vest} = 1$, but the finite measurement tolerance permits a nonzero-width interval. Its small relative width within our adopted range does not alone establish a prior probability, rule out generic UV theories, or require an exact fixed point at unity.

What this does not rule out. Three Phase 6 paths remain open:

1. **Symmetry-driven $\chi_{\rm vest} \to 1$ fixed point.** If a deep-IR symmetry (e.g. Lorentz invariance of the emergent metric sector) makes $\chi_{\rm vest} = 1$ an exact fixed point of the RG flow, the fine-tuning is structural rather than tuned. An exact fixed point at $\chi_{\rm vest} = 1$ would lie inside the compatible interval. The finite-tolerance constraint alone does not select this mechanism or establish that a microscopic model realizes it. Phase 6 should examine whether such a fixed point is realized in the ADW broken phase.

2. **Distinct metric-channel identification.** The "vestigial mode" may not be the second-sound

DOF at all but a separate UV-input metric channel whose susceptibility is naturally pinned to unity. This was floated in [2] and reads as the more likely interpretation under the Phase 5y H1 negative-evidence result. Phase 6e (full nonlinear emergent action) is the natural venue for the disambiguation.

3. **Frequency-dependent $\chi_{\rm vest}$.** Eq. (1) is the leading term in a derivative expansion. If $\chi_{\rm vest}$ varies with frequency, the leading-order constraint could in principle be relaxed by a near-cancellation in the LIGO band. The SK–EFT dispersion correction is the leading frequency dependence in the present framework, and this path is *not* closed — what we can say is conditional. `vestigial_dispersion_below_ligo_at_inspiral_peak` takes $|\gamma| \le 10^{-30}$ on $\Gamma_H/c_{\rm GW}^2$ as a *hypothesis* and proves that under it the correction at the 100 Hz inspiral peak stays inside the GW170817 tolerance. It proves nothing about whether that bound holds: 10^{-30} is a Phase 6a scaffold value, chosen so the correction would sit well below the cap, and it is neither derived nor measured. So a placeholder-sized correction would not relax the constraint; a larger one is not excluded. The general form is `dispersion_within_ligo_iff`, a biconditional exchanging the correction's tolerance for $|\gamma| \le {\rm tol}/\omega$; `dispersion_correction_abs_bound`, `dispersion_correction_linear_in_gamma` and `dispersion_correction_zero_at_no_dissipation` fix its magnitude, additivity and no-dissipation limit. A derivation of Γ_H from a microscopic matching is open.

Comparison to the linearized-EFE wave. The companion linearized-EFE wave [16] ships an emergent-gravity identification *compatible* with observation at the natural-range level: the Sakharov–Adler form $G_{\rm N,emerg} = \alpha_{\rm ADW} \cdot 12\pi/(N_f \Lambda_{\rm UV}^2)$ at $\Lambda_{\rm UV} = M_{\rm Pl}$ produces $\alpha^\star_{\rm ADW} \in [0.40, 1.27]$ for SM N_f, sitting inside the natural range $[0.1, 10]$. The asymmetry between the two waves is a feature of the program: the linearized-EFE $G_{\rm N,emerg}$ depends on the Sakharov–Adler kernel computed at the linearized level, where the relevant scale is $M_{\rm Pl}$; $c_{\rm GW}$ depends on the free metric-channel susceptibility $\chi_{\rm vest}$. Our adopted interval contains unity but derives no value of this susceptibility. The conservative GW170817 cap restricts it to the much narrower compatible interval; the width comparison alone supplies no microscopic selection mechanism.

Reproducibility. The result is reproducible in two layers. The Lean layer is `GravitationalWaves.lean` (32 theorems, zero `sorry`, zero new axioms, zero `maxHeartbeats` overrides) at the project's pinned Mathlib4 commit; the Python layer is `src/gravitational_waves/` with 49 `pytest` cases cross-checking every numeric quoted in the abstract and figure caption. The headline figure (`src/core/visualizations.py:fig_c_GW_vs_ligo_constraint`) and its underlying numerics are committed-pinned.

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